

TITLE PAGE

- Problem Statement ID – SIH26217
- Problem Statement Title- Student Innovation – Renewable/Sustainable Energy
- Theme- Renewable / Sustainable Energy
- PS Category- Hardware
- Team ID- 11
- Team Name- Power IQ

Proposed Solution

- Stationary frame with rotating PV cell rows.
- Single motor synchronizes multiple shafts using a worm-drive.
- Automatic tracking keeps cells aligned with sunlight.

How It Addresses the Problem

- Less moving mass than whole-panel tracking.
- Better daily solar energy harvesting.

Innovation & Uniqueness

- **“Rotate the Cells, Not the Panel.”**
- Single-drive multi-shaft synchronized tracking architecture.
- Designed for compact, energy-efficient and scalable solar tracking.



Hardware

- Solar Cells + Multi-Shaft Worm Gear Mechanism
- NEMA17 Stepper Motor + Stepper Driver
- STM32 Microcontroller
- LDR Sun-Tracking Sensors
- Voltage & Current Sensors

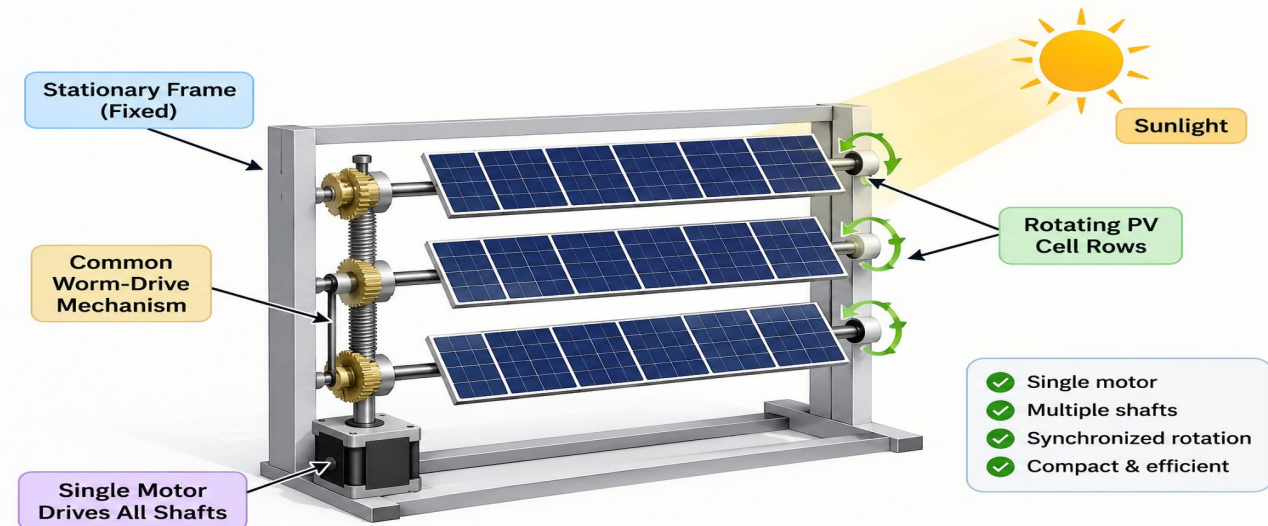
Software & Intelligence

- Embedded C/C++
- IoT Cloud Monitoring
- AI-Based Energy Prediction & Fault Detection

Methodology / Working Flow

Sunlight → LDR Sensors → STM32 → Stepper Motor → Worm Drive → Multi-Shaft Rotation

Voltage/Current Sensors → IoT Cloud → AI Analysis



Proposed Multi-Shaft Solar Cell Tracking Concept

Feasibility

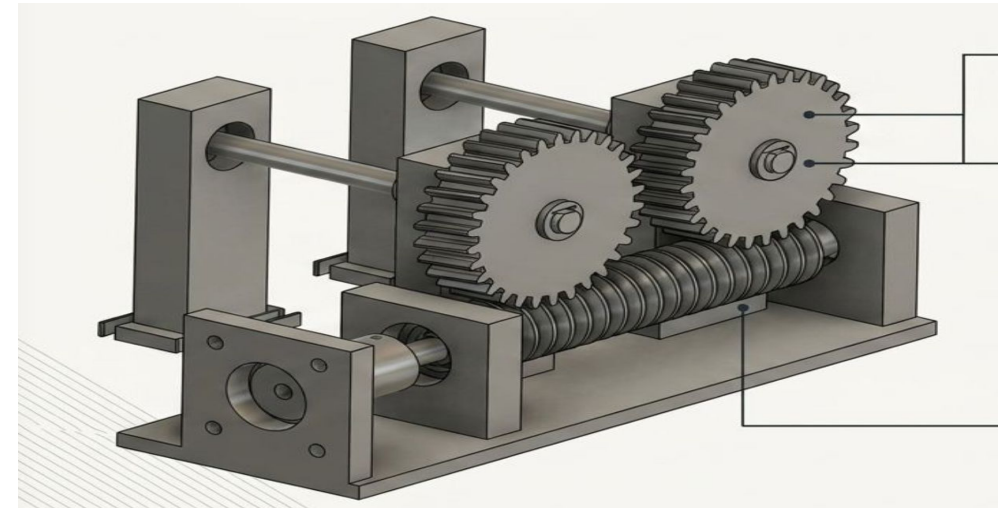
- Single motor operates multiple solar-cell shafts.
- Uses standard, low-cost and easily available components.
- Initial multi-shaft mechanical mechanism successfully tested.
- Modular design allows easy scaling and maintenance.

Challenges & Risks

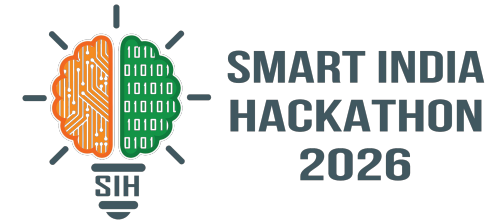
- Gear backlash and shaft synchronization errors.
- Mechanical alignment and increased load with multiple shafts.
- Motor power consumption may reduce net energy gain.
- Outdoor dust, heat and weather may affect reliability.
- Flexible and durable wiring is required because the solar-cell rows continuously rotate during tracking.

Strategies to Overcome

- Precision gear alignment with bearing-supported shafts.
- Step-by-step load and synchronization testing.
- Intermittent tracking to minimize motor energy consumption.
- IoT monitoring for performance and fault detection



IMPACT AND BENEFITS



Potential Impact

• More Solar Energy • Less Space • Simple Installation • Smart Monitoring

- More daily energy from the same solar cells.
- Useful for **On-Grid, Off-Grid, Hybrid and EV Charging** systems.
- Better solar power generation in limited installation space.
- Future use for high-power loads such as **AI & Data Centers**.

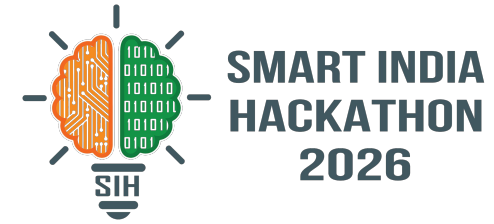
Economic & Practical Benefits

- Higher energy output can reduce the solar capacity needed for the same daily energy demand.
- **Single motor tracks multiple PV cell rows**, reducing the moving mechanism.
- Tracking mechanism can be built **inside the solar module**, making installation simple.
- Ready-made tracking panels can be installed by **regular solar technicians**, without complex tracking structures.

Environmental & Smart Benefits

- More solar energy means less dependence on conventional grid power.
- **IoT**: Live Voltage, Current, Power & Energy monitoring.
- **AI**: Generation prediction, fault detection & maintenance support.

RESEARCH AND REFERENCES



- Details / Links of the reference and research work